SPAM EMAIL CLASSIFICATION

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INTRODUCTION

- EMAIL IS ONE OF THE MOST WIDELY USED COMMUNICATION TOOLS, BUT IT IS OFTEN TARGETED BY SPAM MESSAGES.
- SPAM EMAILS INCLUDE UNWANTED ADVERTISEMENTS, PHISHING LINKS, AND FRAUDULENT CONTENT.
- MANUAL FILTERING IS INEFFICIENT AND TIME-CONSUMING.
- MACHINE LEARNING PROVIDES AN AUTOMATED WAY TO DETECT AND CLASSIFY EMAILS AS SPAM OR HAM BASED ON PATTERNS IN THE DATA.
- THIS PROJECT FOCUSES ON BUILDING AN INTELLIGENT SYSTEM TO IMPROVE EMAIL SECURITY AND FILTERING ACCURACY.

PROBLEM STATEMENT

- Spam emails are increasing rapidly and pose security risks like phishing and fraud.
- Traditional filtering methods are not effective for modern spam.
- Hence, an intelligent Machine Learning-based system is needed to accurately classify emails as Spam or Ham.

OBJECTIVES

- TO DEVELOP A MACHINE LEARNING MODEL FOR CLASSIFYING EMAILS AS SPAM OR HAM
- TO PREPROCESS AND TRANSFORM EMAIL DATA FOR BETTER ANALYSIS
- TO ACHIEVE HIGH ACCURACY WHILE MINIMIZING FALSE POSITIVES AND FALSE NEGATIVES
- TO EVALUATE MODEL PERFORMANCE USING METRICS LIKE ACCURACY, PRECISION, AND RECALL
- TO CREATE AN EFFICIENT AND RELIABLE SPAM DETECTION SYSTEM

DATASET

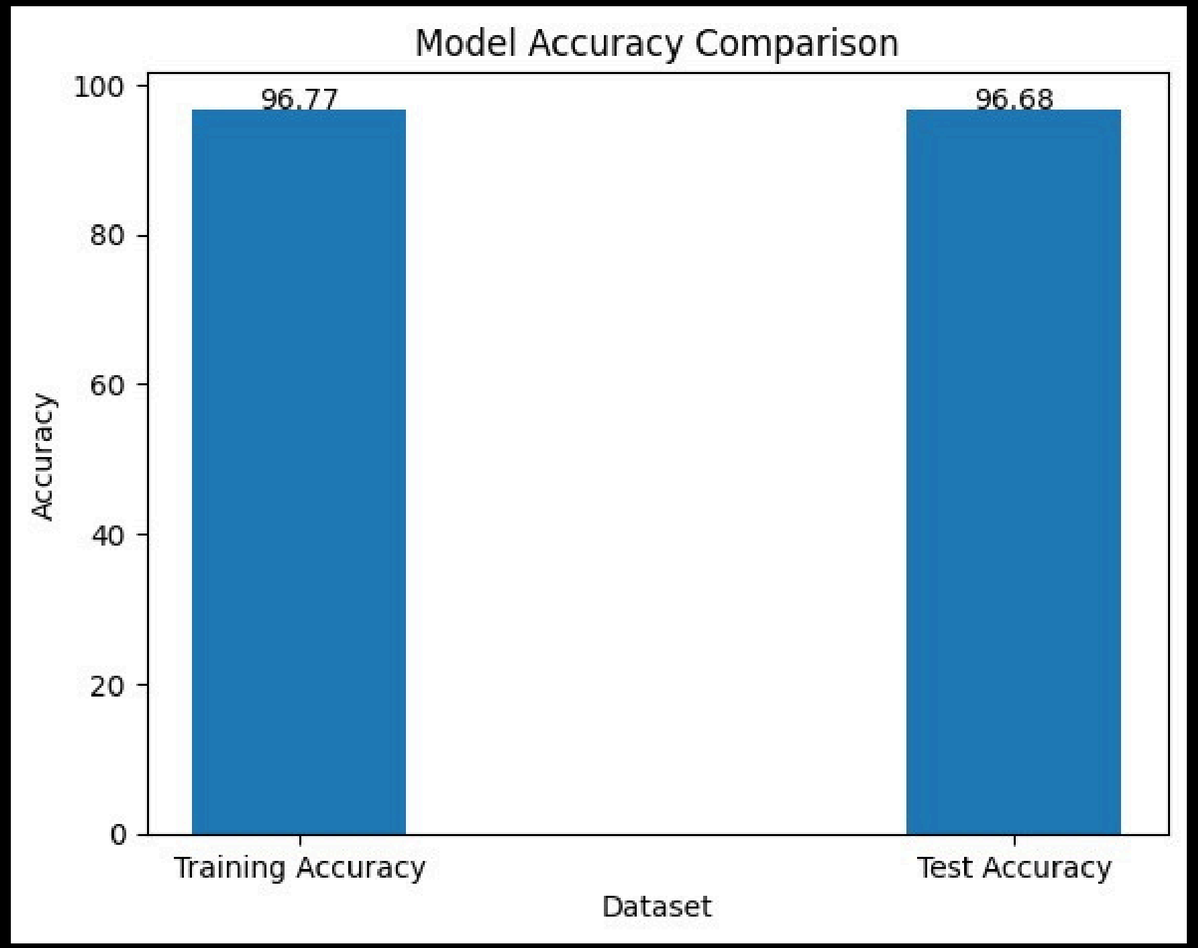
- The dataset used for this project is The Enron Email Dataset (or Kaggle spam dataset)
- It contains a collection of labeled messages classified as Spam or Ham
- Total records: ~5,500 messages
- Features:
  - Message text
  - Label (Spam / Ham)
- The dataset is widely used for training and testing spam classification models

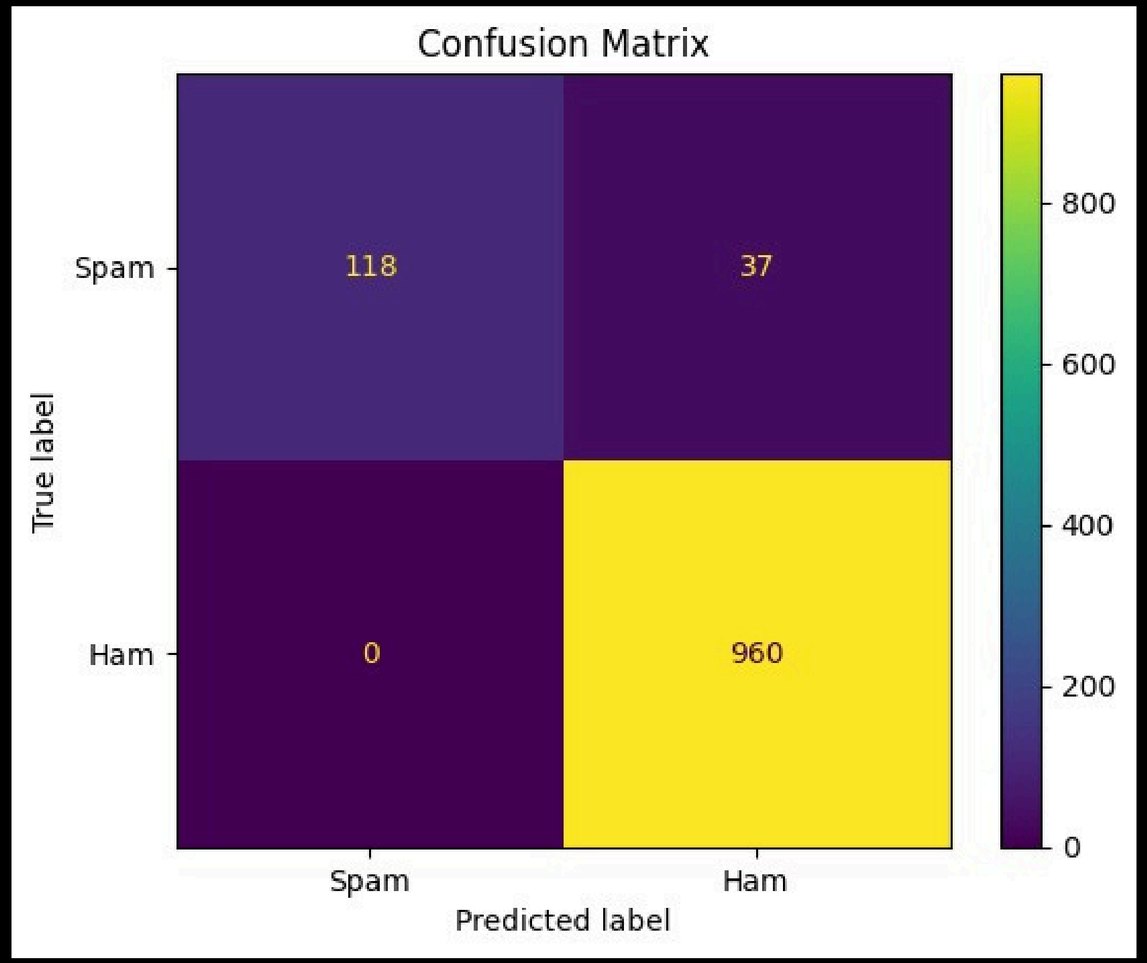
DATA PROCESSING

- Data Cleaning: Remove special characters, numbers, and unwanted symbols
- Lowercasing: Convert all text to lowercase for uniformity
- Tokenization: Split text into individual words
- Stopword Removal: Remove common words like “the”, “is”, “and”
- Stemming/Lemmatization: Reduce words to their root form
- Feature Extraction: Convert text into numerical form using techniques like TF-IDF or Bag of Words

MODEL TRAINING

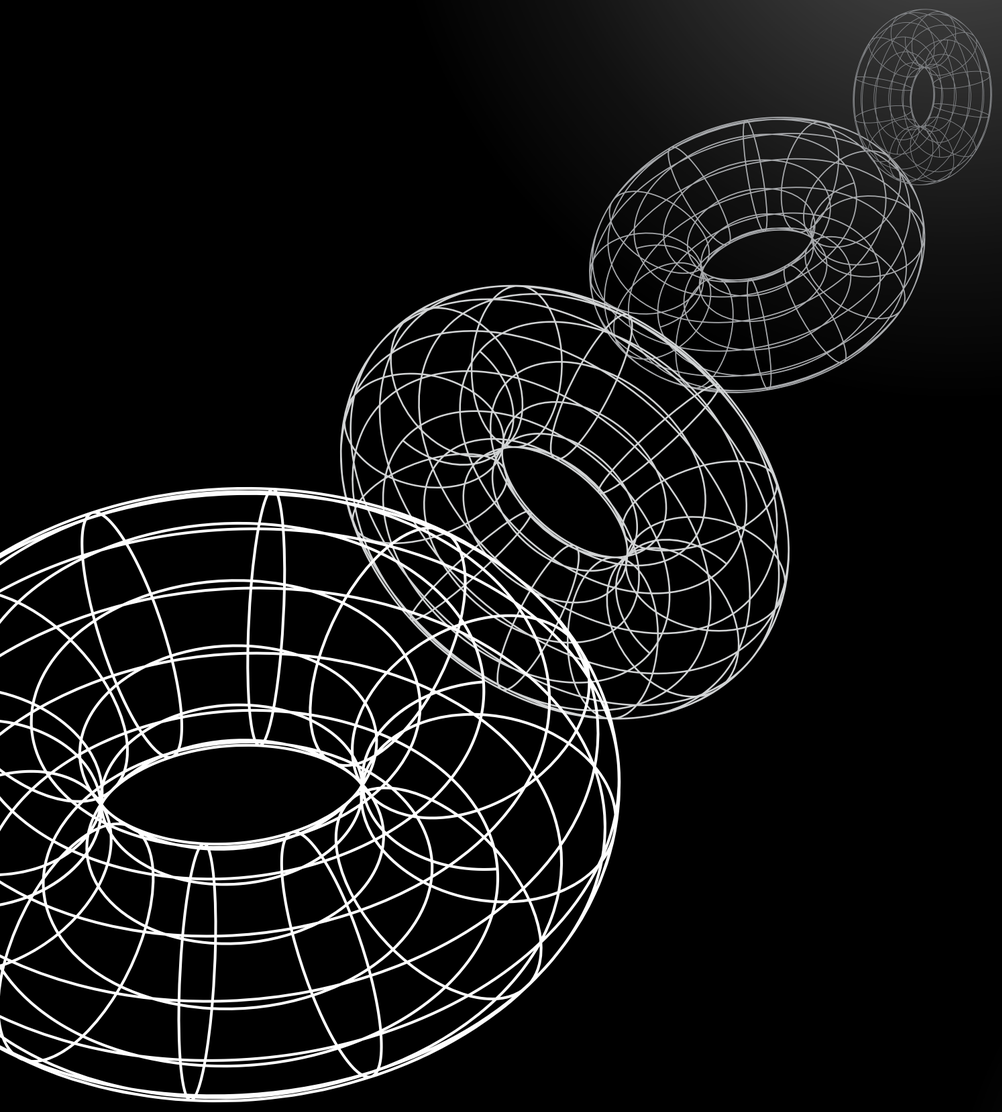
- The dataset is divided into training (80%) and testing (20%) sets.
- The training data is used to teach the model to identify patterns in spam and ham emails.
- Machine Learning algorithms such as Naive Bayes, Logistic Regression, and SVM are applied.
- The model learns from features extracted using techniques like TF-IDF.
- After training, the model is tested on unseen data to evaluate its performance.


Model Accuracy Comparison
100
80
60
40
20
0
Accuracy
96.77
96.68
Training Accuracy
Test Accuracy
Dataset


Confusion Matrix
Spam
Ham
True label
118
37
0
960
Spam
Ham
Predicted label
800
600
400
200
0

MODEL EVALUATION

- The trained model is evaluated using the test dataset
- Accuracy: Measures overall correctness of the model
- Precision: Measures how many predicted spam emails are actually spam
- Recall: Measures how many actual spam emails are correctly detected
- F1-Score: Harmonic mean of precision and recall
- A Confusion Matrix is used to visualize correct and incorrect predictions

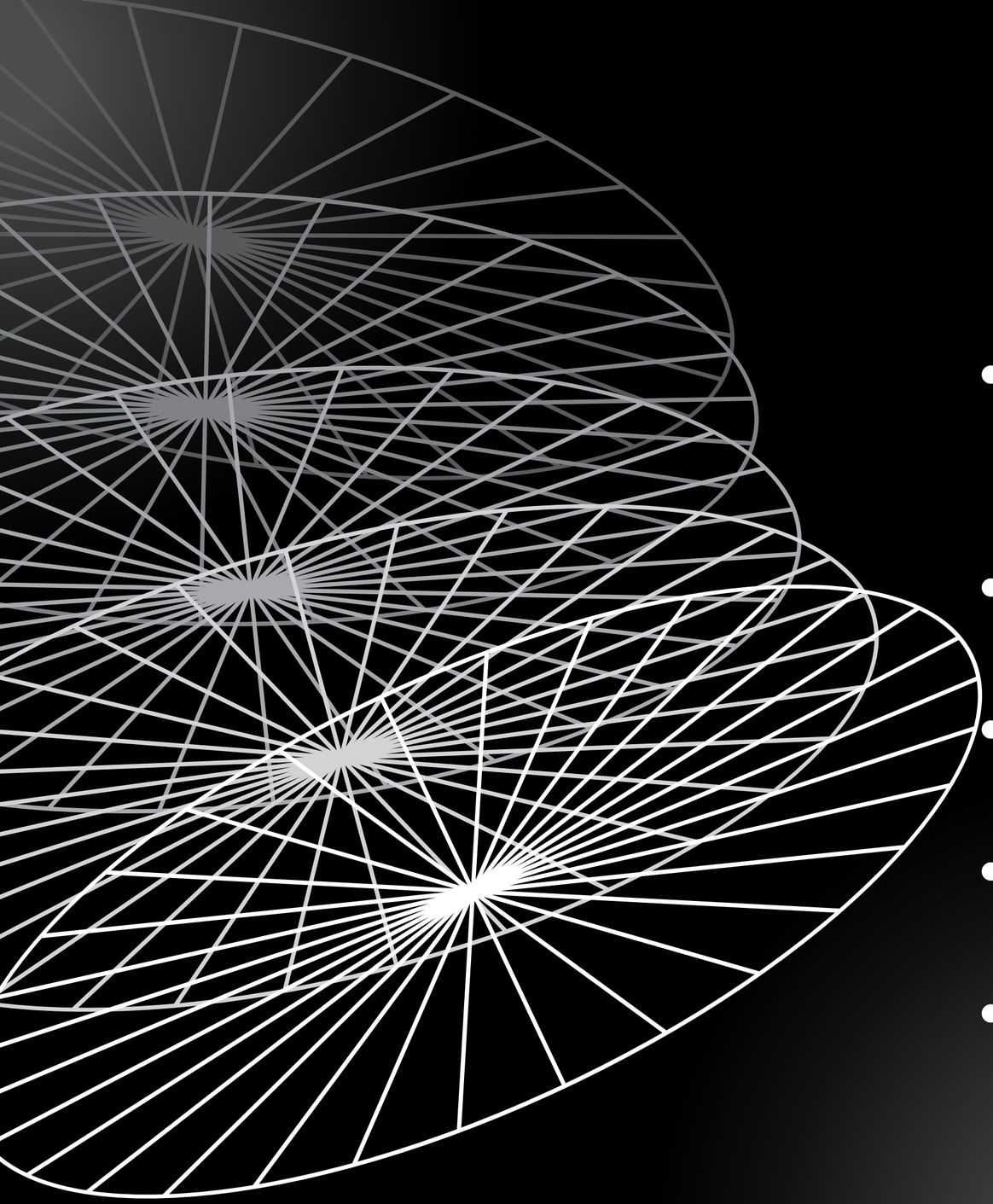


FUTURE SCOPE

- Implement advanced models like Deep Learning (LSTM, BERT) for better accuracy
- Develop a real-time spam detection system
- Integrate the model with email platforms like Gmail or Outlook
- Extend the system to support multiple languages
- Continuously update the model to handle new types of spam

OUTPUT/RESULT

- THE SYSTEM ACCEPTS AN EMAIL OR MESSAGE AS INPUT FROM THE USER
- THE INPUT TEXT IS PREPROCESSED AND TRANSFORMED INTO NUMERICAL FEATURES
- THE TRAINED MACHINE LEARNING MODEL ANALYZES THE INPUT BASED ON LEARNED PATTERNS
- THE SYSTEM CLASSIFIES THE MESSAGE AS SPAM OR HAM (NOT SPAM)
- THE FINAL RESULT IS DISPLAYED CLEARLY TO THE USER
- THE MODEL MAY ALSO PROVIDE A PROBABILITY SCORE INDICATING CONFIDENCE LEVEL
- THIS OUTPUT CAN BE INTEGRATED INTO REAL-WORLD EMAIL SYSTEMS FOR AUTOMATIC SPAM FILTERING



CONCLUSION

- *A Machine Learning-based system for spam email classification was successfully developed*
- *The model effectively classifies emails as Spam or Ham with good accuracy*
- *Data preprocessing and feature extraction play a crucial role in performance*
- *The system helps in improving email security and reducing unwanted messages*
- *This project demonstrates the practical application of ML in real-world problems*

THANK YOU